

WBUHS BIOCHEMISTRY · MODUL

Enzyme Inhibition

Kinetics · Inhibitors · Clinical Applications

SUBJECT

Biochemistry — Paper 1

EXAM RELEVANCE

Group A · B · C · D

PYQ FREQUENCY

10+ times (2010–2025)

SOURCE

KIMS Krishnagar · NMC Bengal PYQ

01 The Big Picture

🔗 WHY THIS MATTERS CLINICALLY

Enzyme inhibition is not just theory — it is the basis of **almost every drug in medicine**. Statins inhibit HMG-CoA reductase. Methotrexate inhibits DHFR. Aspirin irreversibly inhibits COX. Allopurinol is a suicide inhibitor of xanthine oxidase. Understanding inhibition kinetics tells you *how drugs work, why they fail, and how resistance develops*.

02 Types of Inhibition

// CLASSIFICATION FLOWCHART

ENZYME INHIBITION



REVERSIBLE



Competitive

Non-competitive

Uncompetitive

IRREVERSIBLE



Suicide Inhibitors

Heavy Metal Poisoning

Covalent Modification

03

Competitive Inhibition

★ EXAMINER FAV

MECHANISM

// MECHANISM – COMPETITIVE INHIBITOR

Inhibitor structurally similar
to substrate



Binds ACTIVE SITE
of enzyme



Blocks substrate
from binding



↑ [Substrate] can
OVERCOME inhibition



K_m ↑ (apparent)
 V_{max} UNCHANGED

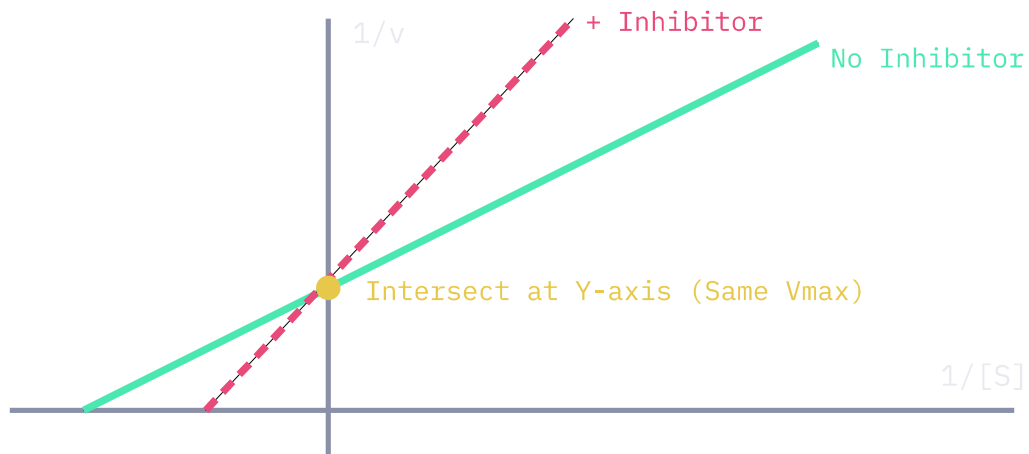


The Golden Rule: In competitive inhibition — **K_m increases** (more substrate needed to achieve half- V_{max}) but **V_{max} remains the same** because high substrate concentration can displace the inhibitor.



Lineweaver-Burk Plot: Lines **intersect on the Y-axis** (same V_{max}). X-intercept shifts toward origin → K_m increased. The two lines must cross the Y-axis at the same point — if they don't, it's NOT competitive.

LINEWEAVER-BURK PLOT: COMPETITIVE



04 Non-Competitive Inhibition

// MECHANISM — NON-COMPETITIVE INHIBITOR

Inhibitor binds
ALLOSTERIC SITE



Binds BOTH free enzyme
AND enzyme-substrate complex



Changes enzyme
conformation



K_m UNCHANGED
 $V_{max} \downarrow \downarrow$

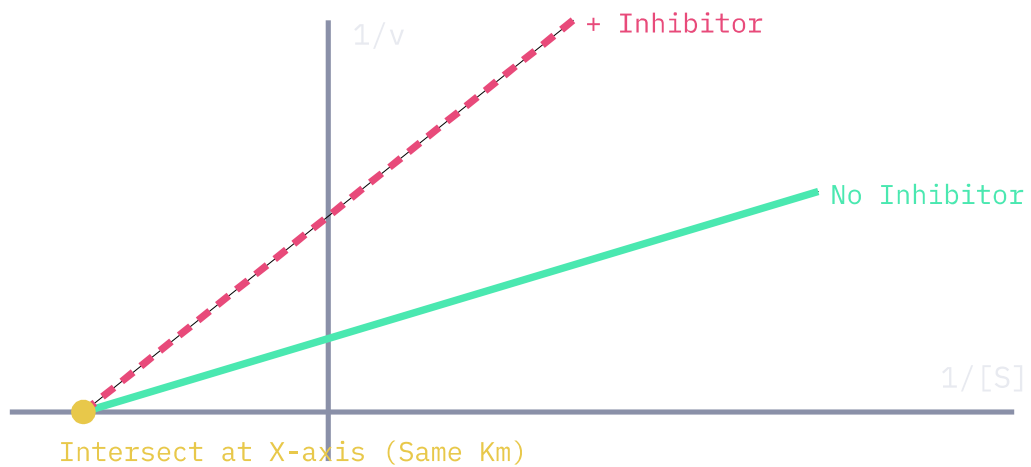


↑ [Substrate] CANNOT overcome —
inhibition is insurmountable

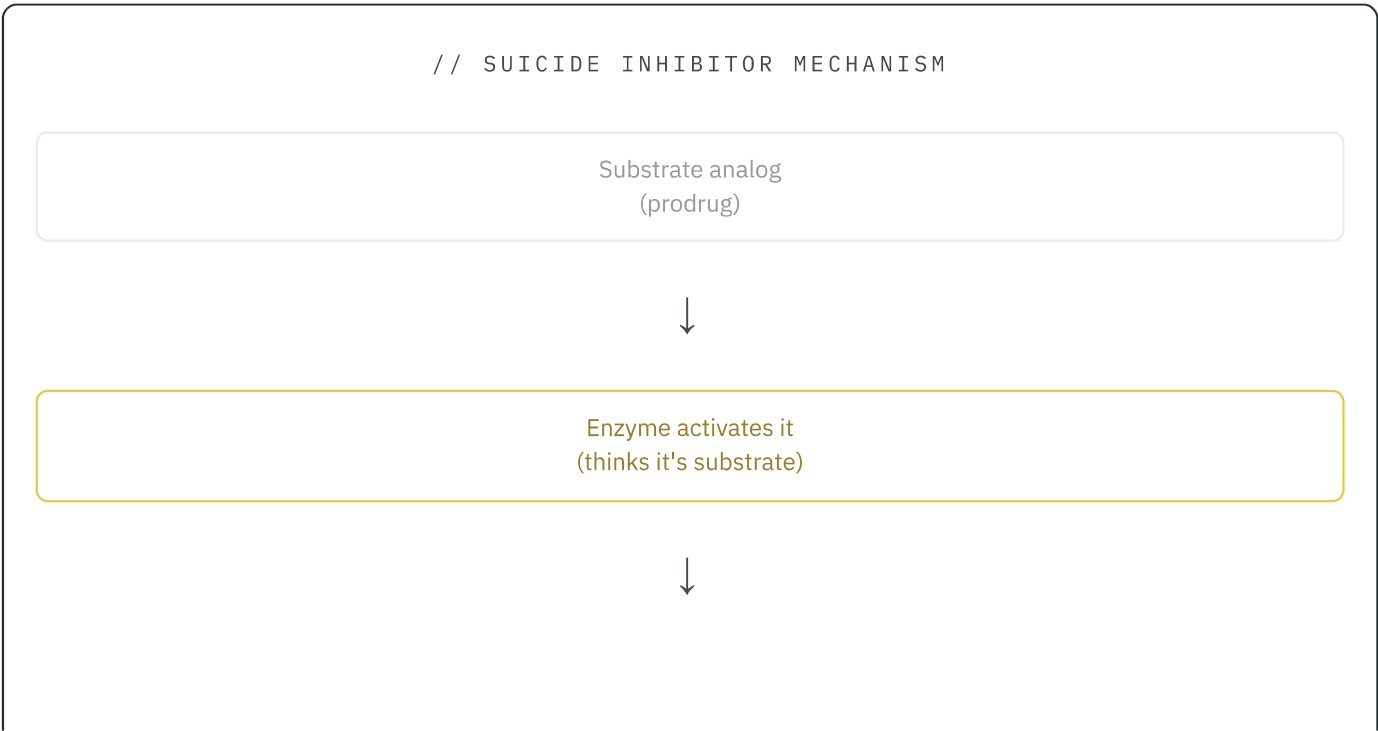


Lineweaver-Burk Plot: Lines **intersect on the X-axis** (same K_m / same $-1/K_m$). Y-intercept increases → V_{max} decreased. Classic: parallel shift? No — they converge at X-axis.

LINEWEAVER-BURK PLOT: NON-COMPETITIVE



PARAMETER	COMPETITIVE	NON-COMPETITIVE
BINDING SITE	Active site	Allosteric site (both E and ES)
EFFECT ON KM	↑ Km (apparent)	No change
EFFECT ON VMAX	No change	↓ Vmax
OVERCOME BY ↑ [S]?	YES ✓	NO ✗
L-B PLOT INTERCEPT	Y-axis (same Vmax)	X-axis (same Km)
KI EXPRESSION	$Km(app) = Km(1 + [I]/Ki)$	$Vmax(app) = Vmax/(1+[I]/Ki)$
CLASSIC EXAMPLE	Methotrexate (DHFR) Statins (HMG-CoA red.) Sulfanilamide (PABA)	Heavy metals (Hg, Pb, As) — specifically bind sulfhydryl (-) Cyanide (Cyt oxidase)



Forms covalent bond
with active site



Enzyme permanently
INACTIVATED 🦋



New enzyme synthesis
needed for recovery

DRUG	ENZYME INHIBITED	CLINICAL USE
ALLOPURINOL	Xanthine oxidase	Gout treatment
ASPIRIN (IRREVERSIBLE)	COX-1 / COX-2	Anti-platelet, anti-inflammatory
PENICILLIN	Transpeptidase (DD-carboxypeptidase)	Antibiotic — cell wall synthesis
EFLORNITHINE	Ornithine decarboxylase	Antiprotozoal (Trypanosomiasis)
5-FLUOROURACIL	Thymidylate synthase	Cancer chemotherapy

07

K_m & V_{max} — The Core Concept

★ EXAMINER FAV

📌 MICHAELIS-MENTEN EQUATION

$$v = (V_{\max} [S]) / (K_m + [S])$$

Where: **v** = reaction velocity, **V_{max}** = maximum velocity, **[S]** = substrate concentration, **K_m** = Michaelis constant



Km Definition: Substrate concentration at which reaction velocity = $\frac{1}{2} V_{max}$. It reflects the *affinity* of enzyme for substrate. **Low Km = High affinity** (less substrate needed). **High Km = Low affinity**.



Hexokinase vs Glucokinase (Classic PYQ!): Hexokinase has **low Km** for glucose — works even at low blood glucose (brain, RBC). Glucokinase has **high Km** — only active after a meal when glucose is high (liver, pancreatic β cells). This is why glucokinase acts as a "glucose sensor."

08

Isoenzymes — LDH & CK

★ EXAMINER FAV

// LDH ISOENZYMES — TISSUE DISTRIBUTION

LDH-1
(H₄)

Heart
RBCs
↑ in MI

LDH-2
(H₃M)

Heart
Kidney
↑ in MI

LDH-3
(H₂M₂)

Lungs
Spleen
Pulmonary

LDH-4
(HM₃)

Kidney
Placenta

LDH-5
(M₄)

Liver
Skeletal muscle
↑ in liver disease



LDH Flipped Pattern in MI: Normally, the concentration of **LDH-2 > LDH-1**. However, during a Myocardial Infarction (MI), massive release from damaged heart tissue causes the ratio to invert, resulting in a "flipped pattern" where **LDH-1 > LDH-2**.

// CK ISOENZYMES

CK-BB

(CK-1)

Brain

CK-MB
(CK-2)

Heart ← MI marker

CK-MM
(CK-3)

Skeletal muscle



CK-MB > 6% of total CK = highly suggestive of Myocardial Infarction. Rises within 4–6 hours, peaks at 12–24 hours. Separated by electrophoresis or immunoassay.

09 Clinical Correlations

DRUG-ENZYME INHIBITION PAIRS (HIGH YIELD)

- ▶ **Methotrexate** → Competitive inhibitor of Dihydrofolate Reductase (DHFR) → Used in cancer, rheumatoid arthritis, psoriasis
- ▶ **Statins** (Atorvastatin, Rosuvastatin) → Competitive inhibitor of HMG-CoA Reductase

RATE LIMITING STEP

 of cholesterol synthesis → Hypolipidemia treatment
- ▶ **Allopurinol** → Suicide inhibitor of Xanthine Oxidase → Reduces uric acid → Gout treatment
- ▶ **Aspirin** → Irreversible inhibitor of COX-1 & COX-2 via acetylation of a specific serine residue at the active site → Anti-platelet effect lasts life of platelet (7–10 days)
- ▶ **Sulfanilamide / Sulfonamides** → Competitive analog of PABA → Inhibits bacterial dihydropteroate synthase → Antibiotic
- ▶ **Penicillin** → Irreversible inhibition of Transpeptidase → Blocks bacterial cell wall synthesis
- ▶ **Neostigmine** → Reversible inhibitor of Acetylcholinesterase → Used in myasthenia gravis
- ▶ **Organophosphates** (insecticides) → Irreversible inhibition of AChE → Toxicity: SLUDGE syndrome

 MNEMONIC 1

"Competitive Kills Km, Non-competitive Nails Vmax"

Competitive → Km changes ($\uparrow$), Vmax stays.

Non-competitive → Vmax changes ($\downarrow$), Km stays.

 MNEMONIC 2 — LINEWEAVER-BURK

"Competitive meets at Y, Non-competitive meets at X"

Competitive inhibition → lines cross at Y-axis intercept (same $1/V_{max}$ = same Vmax).

Non-competitive → lines cross at X-axis intercept (same $-1/K_m$ = same Km).

 MNEMONIC 3 — CK ISOENZYMES

"Brain Beats Muscles" → BB, MB, MM

CK-BB = Brain | CK-MB = Myocardium (Heart) | CK-MM = Muscles

CK-MB is the GOLD STANDARD cardiac marker in early MI (before Troponin era in many exams).

 MNEMONIC 4 — SUICIDE INHIBITORS

"APPLE" — Always Permanently Prohibit Liver Enzymes

Allopurinol (xanthine oxidase) · Penicillin (transpeptidase) · 5-FU (thymidylate synthase) · L-DOPA analog · Eflornithine (ODC)

 VIVA RAPID FIRE — ENZYME INHIBITION

Q1 What happens to Km and Vmax in competitive inhibition?



Km increases (apparent), Vmax remains unchanged. The inhibitor competes with substrate for the active site. High substrate concentration can displace the inhibitor.

Q2 Why is allopurinol called a suicide inhibitor?



Allopurinol is first oxidized by xanthine oxidase to alloxanthine (oxypurinol), which then permanently and covalently binds to the active site of xanthine oxidase, inactivating it irreversibly. The enzyme participates in its own inhibition — hence "suicide."

Q3 Why is K_m of glucokinase higher than hexokinase?

↑

Glucokinase (hepatic) is a low-affinity, high- K_m enzyme that functions as a glucose sensor — it activates only when blood glucose is high (postprandial). Hexokinase has high affinity (low K_m) and works even at basal glucose levels, ensuring glucose supply to brain and RBCs at all times.

Q4 What does the Lineweaver-Burk plot show?

↑

It's a double reciprocal plot of $1/v$ vs $1/[S]$. Y-intercept = $1/V_{max}$. X-intercept = $-1/K_m$. Slope = K_m/V_{max} . Used to distinguish types of inhibition: competitive lines cross at Y-axis; non-competitive lines cross at X-axis.

Q5 Name four non-functional plasma enzymes and their significance.

↑

1) ALT/SGPT — liver damage (hepatocellular). 2) AST/SGOT — liver, cardiac damage. 3) Alkaline Phosphatase — obstructive jaundice, bone disease. 4) Amylase — pancreatitis. 5) CK-MB — Myocardial infarction. These enzymes normally function intracellularly; their appearance in plasma indicates cell damage.

Q6 Differentiate inhibitors from uncouplers (ETC context — frequent combo Q)

↑

Inhibitors block electron transport at specific complexes (e.g., CN^- at Complex IV, Rotenone at Complex I) — stop both electron flow AND ATP synthesis. Uncouplers (e.g., 2,4-DNP, thermogenin) allow electron transport to continue but dissipate the proton gradient — electrons flow but NO ATP is made. Heat is generated instead.

12 Group-D Essay Questions — How to Answer

EQ FORMULA (YOUR TEACHER'S FLOWCHART TRICK)

For every Group-D (Essay/Justify) answer, use this structure:

// EQ ANSWER TEMPLATE — "METHOTREXATE IS A COMPETITIVE ENZYME INHIBITOR"

Define

What is competitive inhibition?



Mechanism Flowchart

MTX → mimics DHFA → binds DHFR active site → blocks tetrahydrofolate synthesis



Kinetics

$K_m \uparrow$, V_{max} unchanged
Overcome by $\uparrow$ substrate



Clinical Consequence

$\downarrow$ THF → $\downarrow$ purine synthesis → $\downarrow$ DNA synthesis → Anti-cancer / Anti-rheumatic



Rescue

Leucovorin (folinic acid) bypasses the block — used to rescue normal cells

WBUHS Biochemistry – Module 01: Enzyme Inhibition

Compiled from PYQ 2010–2025

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